Vish Karthikeyan

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EDUCATION

Stanford University, School of Engineering, California (recipient of the ***Dhirubhai Ambani scholarship***)

Bachelor of Science Degree in Computer Science (Theory of Computation track) | Minor in Public Policy — Exp. June 2027
Tentative Master of Science Degree in Computer Science (Artificial Intelligence track) — Exp. June 2028
Activities and Societies: Director of Outreach: Stanford Political Union ▪ Counselor and Section Leader: Bridge Peer Counseling Center ▪ Teacher: Stanford Splash ▪ Writer: Stanford Political Review ▪ Mentor: Bechtel International Center

CS Academy Erode, India

All India Senior Secondary Certificate, Top percentile national distinction with 96% aggregate (100% in Computer Science) — May 2023
All India Secondary School Exam, Top percentile national distinction with 98% aggregate — May 2021
Activities and Societies: School Captain ▪ National Debate Finalist ▪ National Talent Search Examination Scholar

RELEVANT COURSEWORK

CS 229 (Machine Learning; graduate-level)	CS 124 (Natural Language Processing)	CS 109 (Probability for Computer Scientists)
CS 161 (Design and Analysis of Algorithms)	CS 254 (Complexity Theory; graduate-level)	ECON 50Q and 51 (Quantitative Microeconomics)
CS 106B (Abstract Data Structures)	MATH 104 (Applied Linear Algebra)	CS 107 (Computer Organization and Systems)

SELECTED PROJECTS

Bias Evaluation in LLM vs. Human Criminal Judgments (2025–Present): Matched-guise experiments comparing proprietary LLMs to human professionals on guilt/sentencing decisions using controlled case vignettes; analyzed linguistic cues for bias divergence in probabilistic vs. biological reasoning with Hugging Face Transformers and statistical metrics

Reducing Hallucinations in LLM Coding Agents via Uncertainty-Aware Decisions (2026-Present): Framed tool-calling as uncertainty-guided decision-making; trained agents with supervised fine-tuning and reinforcement learning to defer requests instead of hallucinating invalid APIs/logic, reducing errors in programming tasks using PyTorch, Hugging Face, and uncertainty/entropy signals

Provable Expressivity Limits of Transformer Architectures (2026-Present): Analyzed circuit complexity and language recognition capabilities of transformer encoders/decoders under varying attention mechanisms (hard/soft, unique/averaged) and depths; derived impossibility results via diagonalization and learning theory bounds to explain empirical vs. theoretical performance gaps.

WORK EXPERIENCE

Student Dining Worker, Stanford Residential and Dining Enterprises — Fall 2025 - current

- Hands-on food service operations experience at Stanford's largest dining hall, serving 3000+ people a day in an extremely fast-paced environment
- Helped bridge communication between the student body and dining staff, contributing to a more student-centered campus dining experience

Residential Assistant, Summer Session, Stanford University — Summer 2025

- Built an inclusive, supportive residential community for 300+ high school summer students by planning and leading 3+ programs a week
- Served as a key resource for student concerns and campus referrals; coordinated case management

Teaching Assistant, Frosh 101, Stanford University — Fall 2024

- Inspired community-building conversations and activities amongst two classes of freshmen, shaping their college transitions
- Attended a quarter-long course on inclusive pedagogies and impacts of diversity, inclusion and sense of belonging on learning outcomes

Reunion Homecoming Team Leader, Stanford Alumni Association — Fall 2024

- Led planning efforts, coordinated event logistics, and developed programming for over 7000+ alumni
- Recruited, trained, and managed a team of volunteers, delegating tasks and providing guidance to ensure efficient event execution

Multiple leadership roles, Associated Students of Stanford University — 2023-24

- *Assistant Elections Commissioner:* Facilitated student government elections involving 18,000 voters, supervised data management and security; practiced constitutional hermeneutics and policy development
- *Senate Associate:* Participated in a leadership and advocacy workshop for six months; directed a team working on designing and implementing modifications to Stanford Dining
- *Neighborhood Council Representative:* Managed $200,000 in funding across community building events for 800+ students, engineering various types of events and festivals; designed permanent community spaces for the Stanford Outdoor Enhancement Project

SKILLS AND INTERESTS

Coding	Python (including SciKit-Learn, PyTorch, TensorFlow, NumPy Pandas and Hugging Face), C/C++
Tools	Git and version control, UNIX command-line, SQL (MySQL, SQLite), CSS, HTML, Javascript, RESTful APIs, Docker and containerization, Kubernetes, Agile development, Jupyter, supervised and unsupervised learning, data structures, algorithmic complexity, model evaluation and metrics, feature engineering and data preprocessing
Langauges	English (native proficiency), Tamil (native proficiency), Spanish (elementary) and Hindi (elementary)
Interests	Natural language processing and transformers, large language models, computability and complexity theory, causal inference for public policy, political theory, mental health and peer counseling